Bank Loan Portfolio Analysis

Project Summary | SQL, Excel, and Power BI

Project overview

This project examines a cleaned bank loan dataset containing **65,535 loan records across 52 fields**. The analysis focuses on portfolio size and funding, loan status, repayment performance, and delinquency and default patterns across loan segments. SQL was used for data checks and analysis, Excel for data review and preparation, and Power BI for an interactive two-page report.

Dashboard pages

Portfolio Overview	Loan volume, funded amount, status mix, and risk patterns by grade, term, and state.
Risk & Repayment	Delinquency and default, principal recovery, and payments by state, grade, and loan status; includes Term and Product Code slicers.

Portfolio indicators

Total loans **65,535**	**Total funded amount** **733M**	**Delinquency rate** **10.84%**
Default rate **1.56%**	**Principal recovery rate** **89.12%**	**Total payments** **814.90M**

Amounts are shown in the dataset's displayed monetary units.

Portfolio findings

- Active loans are the largest status group, at approximately **36,000**; fully paid loans total approximately **16,000**.
- Uttar Pradesh has the highest funded amount among the states shown, at approximately **135M**. Punjab and Bihar follow at approximately **116M** and **93M**.
- The portfolio dashboard compares delinquency by grade and default by term, alongside state funding and loan status.

Risk and repayment findings

What the dashboard indicates and how to use the results

Risk patterns

- The delinquency rate rises from **3.92% for Grade A** to **25.95% for Grade G** in the displayed results.
- **Grade E** has the highest displayed default rate at **2.96%**. This is a segment-level observation and should be checked alongside loan counts and other borrower or product characteristics.
- Tripura displays a **100% delinquency rate**. The rate should not be treated as representative until the number of loans in that state is reviewed. A loan-count tooltip is included in the report for this purpose.

Repayment

The dashboard reports total payments of **814.90M** and a principal recovery rate of **89.12%**. The report also breaks payments and principal recovery down by loan status. These comparisons need to account for loan maturity: active loans may still be repaying, so their cumulative payments are not directly comparable with loans that have completed their terms.

Recommended next steps

- **Validate high rates with volume.** Review loan counts for states or segments with unusually high delinquency before prioritizing action.
- **Investigate grade-level risk.** Examine the higher delinquency for Grade G and the higher default rate for Grade E alongside loan terms, products, and borrower characteristics.
- **Separate active and completed loans.** Use status-aware comparisons when assessing repayment and recovery performance.
- **Maintain data-quality checks.** Continue checking category values, missing fields, duplicates, and date or amount anomalies when refreshing the analysis.

Conclusion

The analysis describes a 65,535-loan portfolio with a 10.84% delinquency rate, a 1.56% default rate, and a reported 89.12% principal recovery rate. The two Power BI pages make portfolio performance and risk patterns easier to explore. Segment rates should be interpreted with their loan counts, status, and repayment maturity in view.

Interpretation note
This summary reflects the values shown in the project dashboard and dataset. No reporting date or currency label was provided, so amounts are stated in the dataset's displayed units.